Citation Paraphrase Verification Log

Thesis: *Abstraction Mechanisms for Virtual Avatars*
Nidanur Günay • University of Konstanz • 2026

Entries ordered by first appearance in the thesis (Ch. 1 → Ch. 4).
Check each ☐ once you have opened the original PDF, located the passage, and confirmed the paraphrase is accurate.

ORANGE NOTE = original text not yet verified from PDF. **RED TEXT** = confirmed original sentence.

MANDATORY WORKFLOW RULE — Citation Entry Protocol

Every time a new citation is added anywhere in the thesis, an entry **must** be added to this document before moving on. The entry must include:

- The **exact sentence** from the thesis where the citation appears (copied verbatim).
- The **section** where it appears (e.g. §4.3, §2.1).
- The **original text** from the source PDF (or an ORANGE "NEEDS VERIFICATION" note if the PDF has not yet been read).
- A **notes** field stating whether the citation is correctly matched to the claim.

The ☐ **Validated** checkbox at the bottom of each entry is confirmed by Nidanur after opening the original PDF, locating the passage, and confirming the paraphrase is accurate.

Do not skip this step. A citation added without a corresponding entry here is unverified and must be treated as a potential error until logged.

Chapter 1 — Introduction

#1 nowak2003 — Nowak & Biocca (2003) *Section: §1.1 Motivation*

Claim in thesis: *"Nowak and Biocca demonstrated that avatar appearance exerts a measurable influence on social presence and interpersonal trust."*

Paraphrase used: Paraphrase of Nowak & Biocca's core finding on avatar appearance and social outcomes.

Original text: *"Participants interacting with the less-anthropomorphic image reported more copresence and social presence than those interacting with partners represented by either no image at all or by a highly anthropomorphic image of the other, indicating that the more anthropomorphic images set up higher expectations that lead to reduced presence when these expectations were not met."* □

Notes: Source: `Articles/Nowak2003.pdf`, Abstract. CONFIRMED. Note: the original finding is about copresence and social presence, not trust directly. Verify that the thesis does not overstate "interpersonal trust" if the paper measures only presence/copresence.

Validated — original PDF read; paraphrase confirmed accurate.

#2 weidner2023 — Weidner & Broll (2023) *Section: §1.1 Motivation*

Claim in thesis: *"Weidner and Broll extended this finding to video mediated interaction, showing that the visual design of an avatar functions as a communicative act in its own right."*

Paraphrase used: Paraphrase of Weidner & Broll's finding on avatar fidelity and social perception.

Original text: *"[NEEDS VERIFICATION — open `Articles/Weidner2023.pdf` and locate the sentence about avatar visual design as a communicative act or about video-mediated interaction.]"* □

Notes: PDF available: `Articles/Weidner2023.pdf`. Has not been read. Verify the exact finding and that "communicative act" is supported by the paper.

Validated — original PDF read; paraphrase confirmed accurate.

#3 mori1970 — Mori (1970) *Section: §1.1 Motivation — first use*

Claim in thesis: *"Mori termed this effect the Uncanny Valley, proposing that as a human representation approaches but falls just short of true photorealistic fidelity it tends to evoke unease rather than familiarity."*

Paraphrase used: Paraphrase of Mori's original uncanny valley hypothesis.

Original text: *"I have noticed that, in climbing toward the goal of making robots appear human, our affinity for them increases until we come to a valley (Figure 1), which I call the uncanny valley."* □

Notes: Source: `Articles/uncanny1970.pdf`, p. 2 (IEEE Spectrum translation). CONFIRMED. The thesis paraphrase is accurate. Mori's original is about robots; the thesis correctly extends the concept to digital representations.

Validated — original PDF read; paraphrase confirmed accurate.

#4 mcdonnell2012 — McDonnell et al. (2012) *Section: §1.1 Motivation*

Claim in thesis: *"A claim corroborated empirically for computer generated characters by McDonnell et al."*

Paraphrase used: Citation for empirical corroboration of uncanny valley in CG characters.

Original text: *"[NEEDS VERIFICATION — open `Articles/McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012)...pdf` and locate the central finding about eeriness and render style.]"* □

Notes: PDF available but not yet read. Also cited as `McDonnell09_HappyZombies.pdf` — verify which file is the correct 2012 paper. The thesis cites `mcdonnell2012`.

Validated — original PDF read; paraphrase confirmed accurate.

#5 canales2024 — Canales et al. (2024) *Section: §1.1 Motivation*

Claim in thesis: *"[Uncanny valley claim] corroborated in the specific context of avatar stylisation and trust by Canales et al."*

Paraphrase used: Citation establishing avatar stylisation and trust as the specific research context.

Original text: *"Style differences did not lead to any significant differences in the subjective trust scores, and notably, there was no evidence that the realistic avatars were viewed as more trustworthy."* □

Notes: Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`, §6 Results. CONFIRMED. The citation is appropriate: Canales provides the trust-stylisation context. Verify which exact sentence is closest to the §1.1 use.

Validated — original PDF read; paraphrase confirmed accurate.

#6 gooch1998 — Gooch et al. (1998) *Section: §1.1 Motivation*

Claim in thesis: *"Gooch et al. established a formal model for this approach, replacing physically accurate shading with stylised cues that communicate form without simulating light."*

Paraphrase used: Paraphrase of Gooch's contribution: stylised illumination over physical accuracy.

Original text: *"A non-photorealistic lighting model for automatic technical illustration... uses cool-to-warm color shifts to convey surface orientation without relying on photorealistic shading."* ☐

Notes: Source: `Articles/NPR/Gooch1998.pdf`. CONFIRMED in previous session. The abstract states the model replaces Lambertian diffuse with cool-to-warm transitions. The paraphrase "communicate form without simulating light" is accurate in spirit. Confirm exact wording from abstract.

Validated — original PDF read; paraphrase confirmed accurate.

#7 metasdk — Meta Avatars SDK

Section: §1.2 Problem Statement

Claim in thesis: *"[The SDK] exposing only a single fragment level hook, evaluated after all lighting computation, to third-party developers."*

Paraphrase used: Technical description of the SDK's NPR integration constraint.

Original text: *"[NO PUBLIC PDF — internal SDK documentation. This is an engineering fact verified through implementation, not a citable external source. Confirm the citation is to SDK documentation and not a peer-reviewed paper.]"* ☐

Notes: Internal SDK reference. Cannot be verified from a public PDF. Confirm that the bib entry for `metasdk` correctly cites the SDK documentation with a URL or version number.

Validated — original PDF read; paraphrase confirmed accurate.

#8 isenberg2003 — Isenberg et al. (2003)

Section: §1.2 Problem Statement

Claim in thesis: *"Prior NPR research has not had to address constraints of this kind; the literature, represented by surveys such as Isenberg et al., assumes full pipeline access and custom assets."*

Paraphrase used: Citation for the prior NPR literature assuming full pipeline access.

Original text: *"Generating object silhouettes lies at the heart of nonphotorealism."* ☐

Notes: Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, p. 28. CONFIRMED in previous session. The citation is used here as a representative survey, not for this specific sentence. Verify that the paper indeed assumes full pipeline access (it does — all techniques require custom shader control).

Validated — original PDF read; paraphrase confirmed accurate.

Chapter 2 — Related Work

#9 mori1970 — Mori (1970) *Section: §2.1 Uncanny Valley and Avatar Realism — hypothesis definition*

Claim in thesis: *"The Uncanny Valley hypothesis was first articulated by Mori, who proposed that human affinity toward an artificial figure increases with its degree of human likeness up to a critical threshold, at which point subtle imperfections cause a sharp drop in comfort."*

Paraphrase used: Paraphrase of the core UV hypothesis.

Original text: *"I have noticed that, in climbing toward the goal of making robots appear human, our affinity for them increases until we come to a valley (Figure 1), which I call the uncanny valley."* □

Notes: CONFIRMED. Same source as Entry #3. The thesis paraphrase adds "subtle imperfections cause a sharp drop" which is also supported by Mori's text about the prosthetic hand and zombie examples.

Validated — original PDF read; paraphrase confirmed accurate.

#10 macdorman2006 — MacDorman & Ishiguro (2006) *Section: §2.1 Uncanny Valley — empirically tractable research programme*

Claim in thesis: *"MacDorman and Ishiguro advanced the Uncanny Valley from a qualitative hypothesis into an empirically tractable research programme... they proposed semantic differential instruments to quantify eeriness and strangeness as measurable affective dimensions."*

Paraphrase used: Paraphrase of MacDorman & Ishiguro's methodological contribution.

Original text: *"Mori predicted that, as robots appear more human, they seem more familiar, until a point is reached at which subtle imperfections give a sensation of strangeness. [Additionally:] subtle flaws in appearance and movement can be more apparent and eerie in very humanlike robots."* □

Notes: Source: `Articles/MacDorman2006AndroidScience.pdf`, pp. 299–300. PARTIALLY CONFIRMED. The p. 300 text supports the "sensation of strangeness" claim. HOWEVER: the "semantic differential instruments" claim needs pages beyond p. 300 — read more of the paper to verify this specific contribution. The 4 pages read do not yet show the instrument development.

Validated — original PDF read; paraphrase confirmed accurate.

#11 mcdonnell2012 — McDonnell et al. (2012) *Section: §2.1 Uncanny Valley — render style study*

Claim in thesis: *"McDonnell et al. exposed participants to video stimuli of the same animated character rendered across six distinct styles... Photorealistic rendering was associated with significantly higher eeriness ratings and lower appeal than stylised conditions."*

Paraphrase used: Paraphrase of McDonnell et al.'s key finding on render style and eeriness.

Original text: *"[NEEDS VERIFICATION — read `Articles/McDonnell, R., Breidt, M., and Bülthoff, H. H. (2012)...pdf` and locate the finding about photorealistic rendering and eeriness ratings across style conditions.]"* □

Notes: PDF available. This is the most important citation for the empirical motivation of the thesis. Priority for verification.

Validated — original PDF read; paraphrase confirmed accurate.

#12 seymour2021 — Seymour et al. (2021) *Section: §2.1 Uncanny Valley — field study*

Claim in thesis: *"Seymour et al. compared a near photorealistic digital human to a caricature avatar... Participants rated the photorealistic avatar as more trustworthy; however, the caricature condition did not trigger Uncanny Valley discomfort, occupying instead a stable, neutral region."*

Paraphrase used: Paraphrase of Seymour et al.'s dissociation finding.

Original text: *"[NEEDS VERIFICATION — read `Articles/Seymour, Yuan, Dennis & Riemer, ''Have We Crossed the Uncanny Valley?'' (2021).pdf` and locate the finding about photorealistic vs. caricature trust and eeriness.]"* □

Notes: PDF available. Verify the specific finding about trustworthiness ratings and the caricature not triggering discomfort.

Validated — original PDF read; paraphrase confirmed accurate.

#13 mayer1995 — Mayer et al. (1995) *Section: §2.2 Trust — ABI model definition*

Claim in thesis: *"Mayer et al. defined trustworthiness as comprising three independently evaluable components: ability, benevolence, and integrity."*

Paraphrase used: Paraphrase of the ABI trust model.

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Mayer, Davis, & Schoorman (1995), "An integrative model of organizational trust," Academy of Management Review, 20(3), 709–734. Key sentence likely from Abstract or p. 717.]"* □

Notes: No PDF in `Articles/` folder. This is a foundational paper cited extensively. Obtain the PDF and verify. The ABI model and the definition of trust as "willingness to be vulnerable" are confirmed to be from this paper by the Canales (2024) paper which cites it for the same definition.

Validated — original PDF read; paraphrase confirmed accurate.

#14 mcknight2001 — McKnight & Chervany (2001) *Section: §2.2 Trust — trusting beliefs typology*

Claim in thesis: *"McKnight and Chervany extended this conceptual structure into a typology separating trusting beliefs from trusting intentions, allowing trust measurement to span both attitudinal and behavioural dimensions."*

Paraphrase used: Paraphrase of McKnight & Chervany's typology of trust.

Original text: *"[NEEDS VERIFICATION — read `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and locate the passage about trusting beliefs vs. trusting intentions.]"* ☐

Notes: PDF available: `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf`. Verify this is the correct McKnight & Chervany 2001 paper and confirm the distinction between trusting beliefs and intentions.

Validated — original PDF read; paraphrase confirmed accurate.

#15 alipour2025 — Alipour et al. (2025) *Section: §2.2 Trust — systematic review*

Claim in thesis: *"Alipour et al. conducted a PRISMA-compliant systematic review of 53 empirical studies at the intersection of Uncanny Valley effects and trust."*

Paraphrase used: Paraphrase of Alipour et al.'s scope.

Original text: *"Following PRISMA guidelines, a systematic search identified 53 empirical studies that investigated both UVE-related constructs and trust or trust-related outcomes."* ☐

Notes: Source: `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`, Abstract. CONFIRMED.

Validated — original PDF read; paraphrase confirmed accurate.

#16 alipour2025 — Alipour et al. (2025) *Section: §2.2 Trust — methodological prescription*

Claim in thesis: *"Alipour et al. conclude that well-designed future studies should employ dynamic video stimuli and include at least one behavioural indicator alongside self-report ratings."*

Paraphrase used: Paraphrase of Alipour et al.'s design recommendation.

Original text: *"The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction, and the majority use subjective trust measures."* □

Notes: Source: `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`, Abstract. PARTIALLY CONFIRMED. The abstract confirms the limitation of static images and subjective measures, which implies the recommendation for dynamic stimuli. Read further pages to locate the exact forward-looking prescription sentence.

Validated — original PDF read; paraphrase confirmed accurate.

#17 nowak2005 — Nowak & Rauh (2005) *Section: §2.2 Trust — avatar anthropomorphism and credibility*

Claim in thesis: *"Nowak and Rauh provided an early empirical foundation... perceived anthropomorphism and avatar gender expression both predicted credibility judgments, with more anthropomorphic avatars rated as more credible."*

Paraphrase used: Paraphrase of Nowak & Rauh's credibility finding.

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Nowak & Rauh (2005), "The Influence of the Avatar on Online Perceptions of Anthropomorphism, Androgyny, Credibility, Homophily, and Attraction," Journal of Computer-Mediated Communication.]"* □

Notes: No PDF available. Cannot verify. Confirm the 255 participant count and the finding direction (more anthropomorphic → more credible).

Validated — original PDF read; paraphrase confirmed accurate.

#18 nowak2003 — Nowak & Biocca (2003) *Section: §2.2 Trust — diminishing returns of anthropomorphism*

Claim in thesis: *"Nowak and Biocca extended this framework by demonstrating that anthropomorphism and perceived agency contribute independently to social presence and copresence, and that very high levels of visual anthropomorphism do not consistently produce higher social presence."*

Paraphrase used: Paraphrase of the diminishing-returns finding.

Original text: *"Participants interacting with the less-anthropomorphic image reported more copresence and social presence than those interacting with partners represented by either no image at all or by a highly anthropomorphic image of the other, indicating that the more anthropomorphic images set up higher expectations that lead to reduced presence when these expectations were not met."* ☐

Notes: Source: `Articles/Nowak2003.pdf`, Abstract. CONFIRMED. The abstract directly states the diminishing-returns pattern.

Validated — original PDF read; paraphrase confirmed accurate.

#19 canales2024 — Canales et al. (2024) *Section: §2.2 Trust — multi-component stylisation study*

Claim in thesis: *"Self-reported trust ratings were broadly equivalent across all three style levels; however, the semi-realistic condition attracted the highest rate of behavioural selection."*

Paraphrase used: Paraphrase of Canales et al.'s key dissociation finding.

Original text: *"Trust ratings are largely consistent across the style levels, but participants were more likely to select doctors with the 'Mid' level of stylization for a second opinion."* ☐

Notes: Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`, Abstract. CONFIRMED. Both halves of the dissociation are in the abstract.

Validated — original PDF read; paraphrase confirmed accurate.

#20 gao2025 — Gao et al. (2025) *Section: §2.2 Trust — stylised vs realistic comparable trust*

Claim in thesis: *"Gao et al. found that stylised agents achieved trust levels statistically comparable to realistic agents on all measured dimensions, while voice characteristics were found to have a negligible effect on trust outcomes."*

Paraphrase used: Paraphrase of Gao et al.'s equivalence finding.

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Gao et al. (2025). Verify: (a) trust equivalence across stylisation, (b) voice negligible effect, (c) perceived agent age as moderator.]"* ☐

Notes: No PDF available. The thesis makes three specific claims about this paper (trust equivalence, voice effect, age moderator). All three must be verified.

Validated — original PDF read; paraphrase confirmed accurate.

#21 weidner2023 — Weidner & Broll (2023) *Section: §2.2 Trust — fidelity and self-embodiment*

Claim in thesis: *"Weidner and Broll examined how avatar visual fidelity interacts with self-embodiment and social perception, finding that fidelity effects on social judgments are mediated by the observer's sense of identification with the avatar."*

Paraphrase used: Paraphrase of Weidner & Broll's mediation finding. □

Original text: *"[NEEDS VERIFICATION — read `Articles/Weidner2023.pdf` and locate the mediation finding about identification.]"*

Notes: PDF available. Check the specific claim about mediation by identification. This is distinct from the §1.1 use of the same paper.

Validated — original PDF read; paraphrase confirmed accurate.

#22 bartneck2009 — Bartneck et al. (2009) *Section: §2.2 Trust — Godspeed Questionnaire*

Claim in thesis: *"The Godspeed Questionnaire Series, developed by Bartneck et al., provides validated semantic differential scales for five perceptual dimensions: anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety."*

Paraphrase used: Citation for the Godspeed Questionnaire. □

Original text: *"[NEEDS VERIFICATION — read `Articles/trust/Bartneck_2009_Measurement Instruments for the Anthropomorphism.pdf` and confirm the five subscales and their item descriptions.]"*

Notes: PDF available. Confirm the five subscales. Note: the paper's title is "Measurement Instruments for the Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety of Robots." Confirm this is the Bartneck et al. 2009 Godspeed paper.

Validated — original PDF read; paraphrase confirmed accurate.

#23 macdorman2006 — MacDorman & Ishiguro (2006) *Section: §2.2 Trust — eeriness-specific instruments*

Claim in thesis: *"The eeriness-specific instruments introduced by MacDorman and Ishiguro complement the Godspeed series by targeting the negative affective response to near-human imperfection directly."*

Paraphrase used: Citation for MacDorman's eeriness operationalisation.

Original text: *"[NEEDS VERIFICATION — read remaining pages of `Articles/MacDorman2006AndroidScience.pdf` (read pp. 297–300 so far) to locate the specific eeriness instrument or scale description.]"* □

Notes: Partially read. The 4 pages read do not yet show the scale. Read the rest of the paper. Note: the thesis may be conflating MacDorman 2006 with a different paper for the "eeriness instrument" — verify that the scale is actually presented here vs. in a later MacDorman paper.

Validated — original PDF read; paraphrase confirmed accurate.

#24 ho2010 — Ho & MacDorman (2010)

Section: §2.2 Trust — eeriness and human-likeness orthogonality

Claim in thesis: *"Ho and MacDorman... establishing it as a standalone alternative to the Godspeed eeriness items. Their principal finding is that the two dimensions are orthogonal: the reported correlation between eeriness and human-likeness in their validation data is approximately zero ($r \approx .02$)."*

Paraphrase used: Paraphrase of Ho & MacDorman's orthogonality finding. □

Original text: *"Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)."*

Notes: Source: `Articles/trust/Ho2010UncannyValleyIndices.pdf`, Abstract. CONFIRMED. The $r = .02$ figure is in the abstract. Confirm the instrument description (6 items) by reading further pages.

Validated — original PDF read; paraphrase confirmed accurate.

§2.3 Non-Photorealistic Rendering for Virtual Humans

#25 gooch1998 — Gooch et al. (1998) *Section: §2.3 NPR for virtual humans — cool-to-warm model*

Claim in thesis: *"The foundational contribution to NPR lighting is the cool-to-warm shading model introduced by Gooch et al., which replaced the continuous Lambertian diffuse term with a hue shift from cool shadows to warm highlights."*

Paraphrase used: Paraphrase of Gooch's specific contribution.

Original text: *"A Non-Photorealistic Lighting Model for Automatic Technical Illustration... [replaces] the Lambertian model with cool-to-warm color transitions that convey surface orientation more clearly in line drawings."* □

Notes: Source: `Articles/NPR/Gooch1998.pdf`. CONFIRMED in previous session. Paraphrase accurate. Note: Gooch targets technical illustration, not character rendering — thesis correctly notes this.

Validated — original PDF read; paraphrase confirmed accurate.

#26 lake2000 — Lake et al. (2000) *Section: §2.3 NPR — inverted hull technique*

Claim in thesis: *"The inverted hull technique described by Lake et al. expands backfacing polygons along vertex normals by a small offset, creating a slightly enlarged shell of the mesh."*

Paraphrase used: Paraphrase of Lake et al.'s outline method description.

Original text: *"[PARTIALLY CONFIRMED — Lake et al. (2000) actually describe SED (silhouette edge detection via front/back polygon classification), not the geometry-based inverted hull backface expansion. Read `Articles/NPR/npar2000_lake_et_al.pdf` §4.1 to confirm whether the inverted hull description is accurate for this citation, or whether the citation should be to Isenberg (2003) instead.]"* □

Notes: IMPORTANT FLAG: Previous session confirmed Lake et al. use SED (front/back polygon edge test and line segment rendering), not the vertex-normal expansion described in this sentence. The inverted hull (shell method) may be better attributed to Isenberg 2003. Resolve this before submission.

Validated — original PDF read; paraphrase confirmed accurate.

#27 isenberg2003 — Isenberg et al. (2003) *Section: §2.3 NPR — geometry-based methods incapable of interior*

Claim in thesis: *"Isenberg et al. provide a systematic survey of silhouette algorithms and conclude that geometry-based methods are robust and distance-independent but structurally incapable of detecting internal surface detail."*

Paraphrase used: Paraphrase of Isenberg's survey conclusion on limitations.

Original text: *"Generating object silhouettes lies at the heart of nonphotorealism. [And:] Thus, these geometric methods are not capable of identifying structural boundaries and cannot convey internal surface details."* □

Notes: Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, p. 28. CONFIRMED in previous session. Both the centrality claim and the interior-edge limitation are supported.

Validated — original PDF read; paraphrase confirmed accurate.

#28 gonzalez2018 — Gonzalez & Woods (2018) *Section: §2.3 NPR — Sobel for luminance boundaries*

Claim in thesis: *"The Sobel operator, a discrete gradient approximation applied to the diffuse texture via a 3×3 convolution kernel, identifies luminance boundaries corresponding to clothing patterns, facial contours, and painted surface details."*

Paraphrase used: Citation for Sobel operator and gradient mathematics.

Original text: *"[p. 718, §10.2 "Gradient Operators":] "Obtaining the gradient of an image requires computing the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ at every pixel location in the image." [p. 719, Eqs. (10-24) and (10-25):] $g_x = (z_7+2z_8+z_9)-(z_1+2z_2+z_3)$ and $g_y = (z_3+2z_6+z_9)-(z_1+2z_4+z_7)$ [Sobel kernels with ±2 weighting in centre row/column, as shown in Fig. 10.14.] "These kernels are called the Sobel operators (Sobel [1970])." [p. 719:] "The fact that the Sobel kernels have better noise-suppression (smoothing) characteristics makes them preferable.""* □

Notes: Source: `Articles/Gonzales,Woods-Digital.Image.Processing.4th.Edition.pdf`, pp. 718–719. CONFIRMED. The G_x/G_y matrices with ±2 centre weighting are confirmed in Fig. 10.14 and Eqs. (10-24)–(10-25). NOTE: BT.601 luma coefficients still need separate verification in the colour chapter (Ch. 6). Do not mark the BT.601 claim as confirmed until that is done.

Validated — original PDF read; paraphrase confirmed accurate.

#29 marr1980 — Marr & Hildreth (1980) *Section: §2.3 NPR — low-pass pre-filtering prerequisite*

Claim in thesis:	*"Marr and Hildreth demonstrated that low-pass pre-filtering is a prerequisite for stable gradient computation."*
Paraphrase used:	Citation for the smoothing-before-differentiation principle.
Original text:	*"[NEEDS VERIFICATION — no PDF in collection. Locate Marr & Hildreth (1980), "Theory of Edge Detection," Proceedings of the Royal Society B, 207(1167), 187–217. The key sentence is about Gaussian smoothing before differentiation.]"* □
Notes:	No PDF in collection. Cannot verify directly. The principle is well-established. Confirm the specific claim that Marr & Hildreth use the *isotropy* argument to justify Gaussian (vs. other) smoothing.

Validated — original PDF read; paraphrase confirmed accurate.

#30 barla2006 — Barla et al. (2006) *Section: §2.3 NPR — X-Toon 2D ramp by view angle and depth*

Claim in thesis:	*"Barla et al. introduced X-Toon, which parameterises the shading tone ramp by both view angle and depth, enabling a two-dimensional lookup that encodes artistic intent unavailable in a one-dimensional quantisation step."*
Paraphrase used:	Paraphrase of the X-Toon 2D ramp concept.
Original text:	*"In this paper we describe X-Toon, a toon shader that supports view-dependent effects through two independent extensions to conventional toon shading. The first incorporates a notion of tone detail, so that tone varies with depth or orientation relative to the camera. For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."* □
Notes:	Source: `Articles/NPR/x-toon.pdf`, §1 Introduction (highlighted passage). CONFIRMED in current session. Exact sentence confirmed.

Validated — original PDF read; paraphrase confirmed accurate.

#31 kyprianidis2009 — Kyprianidis et al. (2009) *Section: §2.3 NPR — anisotropic Kuwahara extension*

Claim in thesis: *"Kyprianidis et al. extended the isotropic Kuwahara with an anisotropic formulation aligned to local image structure, producing brush-stroke effects oriented along salient contours; the anisotropic extension requires a structure tensor computed from a prior render pass."*

Paraphrase used: Paraphrase of Kyprianidis's anisotropic extension and its constraint.

Original text: *"We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor. Contrary to conventional edge-preserving filters, our filter generates a painting-like flattening effect along the local feature directions while preserving shape boundaries."* □

Notes: Source: `Articles/NPR/anisotropic_kuwahara.pdf`, Abstract. CONFIRMED in current session. The structure tensor requirement and prior render pass are also confirmed in §3 Method.

Validated — original PDF read; paraphrase confirmed accurate.

#32 praun2001 — Praun et al. (2001) *Section: §2.3 NPR — real-time hatching*

Claim in thesis: *"Praun et al. demonstrated real-time hatching as a complementary branch of this design space, in which tonal regions are conveyed by oriented stroke textures rather than flat fills or edge lines."*

Paraphrase used: Citation for real-time hatching as NPR technique.

Original text: *"[NEEDS VERIFICATION — read `Articles/NPR/hatching.pdf` and confirm it is Praun et al. 2001 and that the description of tonal art maps and oriented stroke textures matches.]"* □

Notes: PDF available. Verify this is the correct paper ("Real-Time Hatching" or "Tonal Art Maps") and confirm the stroke-texture description.

Validated — original PDF read; paraphrase confirmed accurate.

#33 zibrek2019tap — Zibrek et al. (2019, ACM TAP) *Section: §2.3 NPR — social reception study*

Claim in thesis: *"Zibrek et al. conducted a within-participants VR study... Photorealism did not significantly improve emotional recognition or social presence; in some conditions the NPR condition was rated more positively."*

Paraphrase used: Paraphrase of Zibrek et al.'s social reception finding.

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Zibrek et al. (2019), ACM Transactions on Applied Perception.]"* □

Notes: No PDF available. Cannot verify. Confirm the finding direction: NPR $\geq$ photorealism for emotional recognition and social presence.

Validated — original PDF read; paraphrase confirmed accurate.

#34 zibrek2019mig — Zibrek & McDonnell (2019, MIG) *Section: §2.3 NPR — place illusion study*

Claim in thesis: *"Zibrek and McDonnell extended this investigation to embodied VR, finding that photorealism significantly increased place illusion but had a weaker and less consistent effect on social presence."*

Paraphrase used: Paraphrase of Zibrek & McDonnell's dissociation finding. ☐

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Zibrek & McDonnell (2019), MIG 2019.]"*

Notes: No PDF available. Cannot verify. Confirm the dissociation between place illusion and social presence.

Validated — original PDF read; paraphrase confirmed accurate.

#35 latoschik2017 — Latoschik et al. (2017) *Section: §2.3 NPR — abstract/semi-realistic/realistic VR study*

Claim in thesis: *"Latoschik et al. in an immersive social VR context. The semi-realistic condition produced social experiences comparable to the realistic condition on several dimensions, while the abstract condition diverged more substantially."*

Paraphrase used: Paraphrase of Latoschik et al.'s finding on representation levels. ☐

Original text: *"[NEEDS VERIFICATION — read `Articles/Effects of Realism and Representation on Self-Embodied Avatars in Immersive Virtual Environments.pdf` and locate the finding about semi-realistic vs. realistic conditions.]"*

Notes: PDF available. Confirm the three-condition comparison (abstract/semi-realistic/realistic) and that semi-realistic realistic on most social dimensions.

Validated — original PDF read; paraphrase confirmed accurate.

Chapter 3 — Background

#36 gonzalez2018 — Gonzalez & Woods (2018) *Section: §3.2 Edge Detection — Sobel operator*

Claim in thesis: *"The Sobel operator estimates the gradient using two 3×3 kernels: G_x and G_y."*

Paraphrase used: Reference for the Sobel kernel values and gradient magnitude formula.

Original text: *"[p. 719, Fig. 10.14 and Eqs. (10-24)–(10-25):] $g_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$; $g_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$. "These kernels are called the Sobel operators (Sobel [1970])." [p. 716, Eq. (10-17):] "The magnitude, $M(x, y)$, of this gradient vector at a point (x, y) is given by its Euclidean vector norm: $M(x, y) = \|\nabla f(x, y)\| = \sqrt{g_x^2(x, y) + g_y^2(x, y)}$." [p. 720, Eq. (10-26):] Approximate form $M(x, y) \approx |g_x| + |g_y|$ also given."* □

Notes: Source: pp. 716, 719–720. CONFIRMED. The G_x/G_y kernel values in the thesis Eq. (2) match the Gonzalez & Woods matrices exactly (row 3 minus row 1 with ± 2 centre column weighting). The Euclidean magnitude formula and the absolute-value approximation are both confirmed.

Validated — original PDF read; paraphrase confirmed accurate.

#37 marr1980 — Marr & Hildreth (1980) *Section: §3.2.3 Gaussian Pre-Filtering*

Claim in thesis: *"Marr and Hildreth established that stable edge detection requires low-pass smoothing before gradient computation."*

Paraphrase used: Citation for the smoothing-before-differentiation principle in background chapter.

Original text: *"[NEEDS VERIFICATION — same source as Entry #25. No PDF available.]"* □

Notes: Same citation as §2.3. Confirm the Gaussian commutation property (Eq. **??**deriv_commuteeq:gauss_deriv_commuteonsistent with Marr & Hildreth's formulation.

Validated — original PDF read; paraphrase confirmed accurate.

#38 canny1986 — Canny (1986) *Section: §3.2.3 Gaussian Pre-Filtering*

Claim in thesis: *"Canny's influential edge detector built on this principle: smooth with a Gaussian, compute the gradient of the smoothed image, and apply non-maximum suppression to obtain thin, stable edge maps."*

Paraphrase used: Citation for Canny's three-stage edge detection pipeline.

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Canny (1986), "A Computational Approach to Edge Detection," IEEE Transactions on Pattern Analysis and Machine Intelligence, 8(6), 679–698.]"* □

Notes: No PDF available. Confirm the three-stage pipeline (Gaussian smooth → gradient → non-maximum suppression) is accurately described.

Validated — original PDF read; paraphrase confirmed accurate.

#39 kyprianidis2009 — Kyprianidis et al. (2009) *Section: §3.3 The Kuwahara Filter*

Claim in thesis: *"The Kuwahara filter is a structure-preserving smoothing operator that produces piecewise-uniform regions with sharp inter-region boundaries. [Also:] An anisotropic extension by Kyprianidis et al. aligns the filter kernel with local image structure by computing a structure tensor from a prior render pass."*

Paraphrase used: Citation for the Kuwahara filter definition and anisotropic extension.

Original text: *"We present a non-photorealistic rendering technique to transform color images and videos into painterly abstractions. It is based on a generalization of the Kuwahara filter that is adapted to the local shape of features, derived from the smoothed structure tensor."* □

Notes: Source: `Articles/NPR/anisotropic_kuwahara.pdf`, Abstract. CONFIRMED. The isotropic Kuwahara (4 quadrants, minimum variance) is classical prior work; Kyprianidis extends it. The structure tensor requirement is confirmed.

Validated — original PDF read; paraphrase confirmed accurate.

#40 latoschik2017 — Latoschik et al. (2017) *Section: §3.4 Virtual Avatars — representation consequences*

Claim in thesis: *"Where an avatar sits along this continuum has measurable consequences for how users are perceived, how much social presence is felt, and how the avatar is trusted."*

Paraphrase used: Citation for avatar representation level affecting social outcomes (alongside nowak2003). □

Original text: *"[NEEDS VERIFICATION — same source as Entry #31. Read `Articles/Effects of Realism...pdf`.]"*

Notes: Same paper as §2.3. The §3.4 use is for avatar representation continuum effects generally. Confirm this general claim is supported.

Validated — original PDF read; paraphrase confirmed accurate.

#41 yee2007 — Yee & Bailenson (2007) *Section: §3.4 Virtual Avatars — Proteus effect*

Claim in thesis: *"Yee and Bailenson demonstrated through a series of experiments that users assigned more attractive or physically taller avatars adopted behaviours consistent with those characteristics... This tendency, termed the Proteus effect."*

Paraphrase used: Citation for the Proteus effect.

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Yee & Bailenson (2007), "The Proteus Effect: The Effect of Transformed Self-Representation on Behavior," Human Communication Research.]"* □

Notes: No PDF available. Confirm: (a) the specific behaviours mentioned (more disclosure, more assertive negotiation, shorter distances); (b) the term "Proteus effect" is from this paper.

Validated — original PDF read; paraphrase confirmed accurate.

#42 ho2010 — Ho & MacDorman (2010) *Section: §3.5 Trust and Social Presence — eeriness instrument, orthogonality*

Claim in thesis: *"Ho and MacDorman developed a separate eeriness instrument... Their central methodological finding is that eeriness and human-likeness are orthogonal perceptual dimensions: the reported correlation between the two dimensions in their validation data is approximately zero ($r \approx .02$)."*

Paraphrase used: Paraphrase of Ho & MacDorman's key methodological contribution. □

Original text: *"Phase 2 of this study developed alternative indices with non-significant correlations ($p > .05$) between the proposed y-axis eeriness and x-axis perceived humanness ($r = .02$)."*

Notes: CONFIRMED — same source as Entry #20. Exact $r = .02$ figure confirmed from Abstract.

Validated — original PDF read; paraphrase confirmed accurate.

#43 alipour2025 — Alipour et al. (2025) *Section: §3.5 Trust — systematic review, negative association*

Claim in thesis: *"A systematic review by Alipour et al. examined the relationship between uncanny valley discomfort and trust across 42 empirical studies. Their review found a consistent negative association... Alipour et al. also found that the association is strongest when stimuli are presented as dynamic video clips."*

Paraphrase used: Paraphrase of Alipour et al.'s meta-analytic finding on eeriness and trust.

Original text: *"The results of our systematic review reveal that most studies rely on static images or hypothetical scenarios with limited real-time interaction."* ☐

Notes: Source: `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`, Abstract. PARTIALLY CONFIRMED. The abstract confirms the static image dominance. The specific claim about "42 studies" and "consistent negative association" and "strongest for video stimuli" must be verified from the body of the paper (pages 4+).

Validated — original PDF read; paraphrase confirmed accurate.

Chapter 4 — Methodology

#44 isenberg2003 — Isenberg et al. (2003) *Section: §4.3 NPR Technique Design — opening claim*

Claim in thesis: *"NPR techniques in computer graphics encompass a wide range of abstraction levels."*

Paraphrase used: General NPR survey citation supporting the range-of-abstraction claim. Swapped from gooch1998 (2025-07-16): Gooch (1998) is a single-technique paper on cool-to-warm lighting, not a survey of abstraction levels. Isenberg (2003) surveys multiple NPR technique classes and is the closest general NPR survey in the bib. ☐

Original text: *"Generating object silhouettes lies at the heart of nonphotorealism."*

Notes: Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, p. 28. CONFIRMED in previous session. Note: the opening claim is broader than what Isenberg explicitly states (he focuses on silhouettes). If a comprehensive NPR survey is added to the bib in future, replace this citation. ☐ re-check at submission.

Validated — original PDF read; paraphrase confirmed accurate.

#45 zell2015 — Zell et al. (2015) *Section: §4.3 NPR Technique Design — excessive stylisation*

Claim in thesis: *"Choosing a level that is too low may place an avatar within the uncanny valley, whereas excessive stylisation can reduce visual detail and compromise expressive readability."*

Paraphrase used: Citation for the claim that excessive stylisation degrades expressive readability.

Original text: *"The perceived intensity of expressions decreases with realism of shape, but is nearly independent of material stylization."* □

Notes: Source: Zell et al. (2015), Main Findings list, p. 184:2. CONFIRMED in previous session. The finding direction: as shape becomes more stylised (less realistic), perceived expression intensity decreases. This supports "excessive stylisation undermines expressive readability."

Validated — original PDF read; paraphrase confirmed accurate.

#46 wallraven2008 — Wallraven et al. (2008) *Section: §4.3 NPR Technique Design — excessive stylisation (paired with zell2015)*

Claim in thesis: *"[Same sentence as above, paired citation with zell2015:] excessive stylisation can reduce visual detail and compromise expressive readability."*

Paraphrase used: Citation for shape detail reduction degrading perceived expression quality.

Original text: *"In addition, in Experiment 2, we found clear effects of shape blurring on both intensity and sincerity judgments. This demonstrates that, even though reduced shape information does not affect recognition, it does influence the perceived quality of the expression."* □

Notes: Source: Wallraven et al. (2008), §4 Summary, p. 23:19. CONFIRMED in previous session.

Validated — original PDF read; paraphrase confirmed accurate.

#47 isenberg2003 — Isenberg et al. (2003) *Section: §4.3.2 V1 Inverted Hull — silhouettes as NPR foundation*

Claim in thesis: *"Object silhouettes are a defining visual feature of nonphotorealistic rendering."*

Paraphrase used: Opening claim for V1 section.

Original text: *"Generating object silhouettes lies at the heart of nonphotorealism."* □

Notes: Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, p. 28. CONFIRMED in previous session.

Validated — original PDF read; paraphrase confirmed accurate.

#48 isenberg2003 — Isenberg et al. (2003) *Section: §4.3.2 V1 Inverted Hull — stable silhouette without screen-space processing*

Claim in thesis: *"The inverted hull method was selected because it produces stable geometric silhouettes without requiring screen space processing."*

Paraphrase used: Selection rationale for V1.

Original text: *"[NEEDS CONFIRMATION of the exact sentence. Isenberg describes geometry-based methods as "robust and distance-independent." Locate the exact sentence about screen-space independence in `Isenberg_2003_ADG.pdf`.]"* ☐

Notes: Confirmed in previous session that Isenberg describes geometry-based methods as robust and distance-independent. Verify exact wording for "without requiring screen space processing."

Validated — original PDF read; paraphrase confirmed accurate.

#49 isenberg2003 — Isenberg et al. (2003) *Section: §4.3.2 V1 — vertex stage, predictable cost*

Claim in thesis: *"This approach is well suited to real time NPR avatar rendering because it requires no offline mesh preprocessing and executes entirely within the GPU vertex transformation stage."*

Paraphrase used: Technical justification for V1 selection.

Original text: *"[NEEDS VERIFICATION — locate in Isenberg (2003) the sentence describing geometry-based methods as operating at the vertex stage without preprocessing. Read `Isenberg_2003_ADG.pdf`.]"* ☐

Notes: Partially confirmed from previous session. Verify exact passage.

Validated — original PDF read; paraphrase confirmed accurate.

#50 isenberg2003 — Isenberg et al. (2003) *Section: §4.3.2 V1 — limitation: only external silhouette*

Claim in thesis: *"Creases between body regions, clothing seams, and facial contours receive no edge marking, as the method captures only the external silhouette of the mesh."*

Paraphrase used: Paraphrase of the interior-edge limitation.

Original text: *"Thus, these geometric methods are not capable of identifying structural boundaries and cannot convey internal surface details."* ☐

Notes: Source: `Articles/NPR/Isenberg_2003_ADG.pdf`. CONFIRMED in previous session. Also supported by Lake et al.: "in addition to silhouettes, we find edge lines that mark other key features of a model, such as crease and border edges" (implying silhouette-only methods miss these).

Validated — original PDF read; paraphrase confirmed accurate.

#51 gooch1998 — Gooch et al. (1998) *Section: §4.3.3 V2 Toon Shading — [6] stylised lighting models preserve surface orientation*

Claim in thesis: *"Previous work has shown that stylised lighting models can preserve the perception of surface orientation while departing from physically based illumination [6]."*

Paraphrase used: Numeric citation [6] = gooch1998 in compiled PDF.

Original text: *"[The cool-to-warm model] uses both luminance and changes in hue to indicate surface orientation... [replacing] traditional illumination models [that] do not show surface orientation well in areas that are not brightly lit."* □

Notes: Source: `Articles/NPR/Gooch1998.pdf`. CONFIRMED in previous session. Gooch's model was explicitly designed to communicate surface orientation through colour rather than physical illumination.

Validated — original PDF read; paraphrase confirmed accurate.

#52 lake2000 — Lake et al. (2000) *Section: §4.3.3 V2 Toon Shading — [9] discretising illumination for flat appearance*

Claim in thesis: *"[D]iscretising the illumination response into uniform tonal regions produces the characteristic flat appearance associated with hand drawn animation [9]."*

Paraphrase used: Numeric citation [9] = lake2000 in compiled PDF.

Original text: *"Rather than smoothly interpolating shading across a surface, our hard shading technique finds a transition boundary and shades each side of the boundary with a solid color."* □

Notes: Source: `Articles/NPR/npar2000_lake_et_al.pdf`. CONFIRMED in previous session. Lake's "hard shading technique" is exactly the discrete tonal approach described.

Validated — original PDF read; paraphrase confirmed accurate.

#53 lake2000 — Lake et al. (2000) *Section: §4.3.3 V2 Toon Shading — quantisation formula*

Claim in thesis: *"This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading."*

Paraphrase used: Citation for the floor quantisation formula $t = \lfloor s \cdot n \rfloor / n$.

Original text: *"Rather than smoothly interpolating shading across a surface, our hard shading technique finds a transition boundary and shades each side of the boundary with a solid color."* □

Notes: Source: `Articles/NPR/npar2000_lake_et_al.pdf`. CONFIRMED. Note: Lake et al. use a 1D texture lookup, not an explicit floor function. The floor formula in the thesis is a direct implementation equivalent. The citation is defensible as the conceptual source.

Validated — original PDF read; paraphrase confirmed accurate.

#54 barla2006 — Barla et al. (2006) *Section: §4.3.4 V2.2 X-Toon — 2D ramp replacing 1D*

Claim in thesis: *"Barla et al. propose replacing the one-dimensional ramp with a two-dimensional ramp texture, where the horizontal axis encodes the lighting response and the vertical axis encodes the abstraction level."*

Paraphrase used: Paraphrase of the 2D ramp formulation.

Original text: *"For this, we replace the conventional 1D texture used in toon shading with a 2D texture, where the second dimension corresponds to desired tone 'detail'."*

Notes: Source: `Articles/NPR/x-toon.pdf`, §1 Introduction. CONFIRMED in current session. The axis assignments are confirmed in §3 Tone Detail.

Validated — original PDF read; paraphrase confirmed accurate.

#55 barla2006 — Barla et al. (2006) *Section: §4.3.4 V2.2 X-Toon — Normal Field Abstraction*

Claim in thesis: *"Barla et al. term this technique Normal Field Abstraction, achieved by blending the original mesh normals with normals derived from a geometrically simplified shape."*

Paraphrase used: Citation for the shape detail / normal field technique.

Original text: *"We achieve this by modifying the surface normal field used in shading computations with a geometric map: we map the input mesh to an abstracted shape that acts as a lower bound for shape detail. [The designer controls] the "abstraction" of shape detail by setting a blending parameter that controls how the shader interpolates between normals of the input mesh and the abstracted shape."*

Notes: **CRITICAL FLAG:** Barla et al. do NOT use the term "Normal Field Abstraction" anywhere in the paper. The technique is described under Section 4 "Shape detail." The thesis sentence "Barla et al. term this technique Normal Field Abstraction" is INCORRECT — they do not give it this name. Revise the thesis sentence to: "Barla et al. describe a shape detail extension that modifies the surface normal field by blending original mesh normals with normals from a geometrically abstracted shape [**?**]." Source: `Articles/NPR/x-toon.pdf`, §4.

Validated — original PDF read; paraphrase confirmed accurate.

#56 isenberg2003 — Isenberg et al. (2003) *Section: §4.3.3 V3 — image space taxonomy opening claim*

Claim in thesis: *"Normal based edge detection belongs to the class of image space NPR techniques, in which per pixel surface properties are analysed to reveal geometric boundaries."*

Paraphrase used: Taxonomic claim: image space vs. object space NPR classification.

Original text: *"[Classification section, p. 29:] "we distinguish between image space algorithms that only operate on image buffers, hybrid algorithms that perform manipulations in object space but yield the silhouette in an image buffer, and object space algorithms that perform all calculations in object space." [Image space algorithms section, p. 29:] "Image space algorithms exploit discontinuities in the image buffer(s) that result from conventional rendering and extract them using image-processing methods.""*

Notes: Source: `Articles/NPR/Isenberg_2003_ADG.pdf`, pp. 29–30. CONFIRMED. Both the three-way classification and the image-processing definition are directly stated. The phrase "per pixel surface properties are analysed" is a paraphrase of the image buffer discontinuity detection approach.

Validated — original PDF read; paraphrase confirmed accurate.

#57 saitotakahashi1990 — Saito & Takahashi (1990) *Section: §4.3.3 V3 — G-buffer normal storage for edge detection*

Claim in thesis: *"The underlying principle of normal based edge detection originates from image space NPR techniques. Saito and Takahashi introduced the concept of storing per pixel surface normals in geometric buffers and detecting edges from discontinuities in these buffers."*

Paraphrase used: Citation for the G-buffer normal vector storage concept.

Original text: *"[Abstract, confirmed from SIGGRAPH History page:] "data about geometric properties of surfaces are preserved as Geometric Buffers (G-buffers), with each G-buffer containing one geometric property such as the depth or the normal vector of each pixel" and the algorithm detects "discontinuities, edges, contour lines, and curved hatching.""*

Notes: Source: `Articles/NPR/Comprehensible Rendering of 3-D Shapes Saito.pdf` — physical PDF read directly (SIGGRAPH 1990, pp. 197–200). CONFIRMED from physical source. Key sentences: [p. 198, §2:] "Each buffer contains one geometric property, such as the depth or **the normal vector** of the visible object in each pixel." G-buffer property list explicitly includes "nx: normal vector x / ny: normal vector y / nz: normal vector z". [§3, p. 198:] "We develop drawing algorithms for the basic enhancement operations, the drawing of discontinuity lines, contour lines, and curved hatching. All of them are realized with 2-D image processing operations." The thesis claim "Per-pixel surface normals stored in G-buffer channels have been used for edge detection" is precisely and fully supported.

Validated — original PDF read; paraphrase confirmed accurate.

#58 nienhaus2003 — Nienhaus & Döllner (2003) *Section: §4.3.3 V3 — normal buffer discontinuities detect silhouette, border, and crease edges*

Claim in thesis: *"Building on this idea, Nienhaus and Döllner demonstrated that discontinuities in the normal field provide a reliable cue for silhouette, border, and crease edges in real time NPR rendering."*

Paraphrase used: Paraphrase of the edge classification and normal-buffer detection method.

Original text: *"[Abstract, p. 1:] "It is based on the edge map, a 2D texture that encodes visually important edges of 3D scene objects, and includes silhouette edges, border edges, and crease edges." [Section 3, p. 3:] "A silhouette edge is an edge adjacent to a polygon facing towards the camera (front-facing) and one polygon facing in the opposite direction (back-facing). A border edge is an edge to exactly one polygon. A crease edge is an edge between two front-facing (or back-facing, respectively) polygons whose dihedral angle is above some threshold." [Section 4.1.2, p. 3–4:] "Abrupt changes in the normal-buffer occur typically at crease edges... 0th order discontinuities of the normal-buffer indicate crease edges and, furthermore, profile edges.""* □

Notes: Source: `Articles/NPR/Edge Enhancement.pdf` — physical PDF read. CONFIRMED. Key original sentences: [Abstract, p. 1:] "It is based on the edge map, a 2D texture that encodes visually important edges of 3D scene objects, and includes silhouette edges, border edges, and crease edges." [§3, p. 3 — three edge type definitions confirmed verbatim.] [§3, p. 3:] "In a typical 3D scene, abrupt changes in the z-buffer occur at silhouette edges and border edges." [§3, p. 3:] "**Abrupt changes in the normal-buffer occur typically at crease edges.**" **PRECISION FLAG RESOLVED:** The current thesis text at §4.3.3 already satisfies the precision requirement: it says "G-buffer channels" (plural, covering both z-buffer and normal buffer) and explicitly qualifies that crease edges are "specifically recovered from the normal buffer." The text does NOT claim all three edge types come from the normal buffer alone. No revision needed.

Validated — original PDF read; paraphrase confirmed accurate.

#59 gonzalez2018 — Gonzalez & Woods (2018) *Section: §4.3.5 V4 Sobel — Sobel kernels and gradient magnitude*

Claim in thesis: *"The horizontal and vertical gradient components G_x and G_y are estimated using the Sobel kernels defined in Equation (2). Rather than detecting edges from discontinuities in the surface normal field, it computes image gradients directly from the avatar texture using the Sobel operator."*

Paraphrase used: Reference to Gonzalez & Woods for Sobel kernel values.

Original text: *"[p. 718:] "Obtaining the gradient of an image requires computing the partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ at every pixel location in the image." [p. 719:] "These kernels are called the Sobel operators (Sobel [1970])." [p. 720:] "Any of the pairs of kernels from Fig. 10.14 are convolved with an image to obtain the gradient components g_x and g_y at every pixel location. These two partial derivative arrays are then used to estimate edge strength and direction.""* □

Notes: Source: pp. 718–720. CONFIRMED. Same kernel definition confirmed as Entry #24 and #33. The sentence "computes image gradients directly from the avatar texture" is precisely supported by "computing the partial derivatives at every pixel location in the image" (p. 718). The phrase "using the Sobel operator" is directly supported by the named operator definition on p. 719.

Validated — original PDF read; paraphrase confirmed accurate.

#60 canny1986 — Canny (1986) *Section: §4.3.5 V4 Sobel — gradient detectors require prior smoothing*

Claim in thesis: *"V4.2 addresses this by pre-smoothing the texture input before differentiation, following Canny's established principle that gradient-based detectors require prior smoothing to suppress noise sensitivity."*

Paraphrase used: Citation for the smoothing motivation. □

Original text: *"[NEEDS VERIFICATION — same source as Entry #34. No PDF available. Confirm Canny states this principle explicitly.]"*

Notes: No PDF. The principle is standard; confirm Canny is the correct citation vs. Marr & Hildreth (1980), who make the same argument.

Validated — original PDF read; paraphrase confirmed accurate.

#61 marr1980 — Marr & Hildreth (1980) *Section: §4.3.6 V4.2 Gaussian Sobel — Gaussian is isotropic, no orientation bias*

Claim in thesis: *"The Gaussian filter is the preferred smoothing choice because it is isotropic: it attenuates noise equally in all spatial directions without introducing an orientation bias that would systematically skew the distribution of detected edge directions."*

Paraphrase used: Paraphrase of Marr & Hildreth's justification for Gaussian pre-filtering. □

Original text: *"[NEEDS VERIFICATION — same source as Entry #25. No PDF. Confirm Marr & Hildreth use the isotropy argument specifically.]"*

Notes: No PDF. The thesis makes a specific claim about the isotropy argument. Verify this is from Marr & Hildreth and not from another source such as Canny.

Validated — original PDF read; paraphrase confirmed accurate.

#62 gonzalez2018 — Gonzalez & Woods (2018) *Section: §4.3.6 V4.2 Gaussian — classical 3×3 discrete approximation*

Claim in thesis: *"The defaults follow the classical discrete 3×3 Gaussian approximation: centre $= 0.25$, cardinal $= 0.125$, diagonal $= 0.0625$."*

Paraphrase used: Reference to Gonzalez & Woods for the standard Gaussian kernel coefficients. □

Original text: *"[NEEDS VERIFICATION — same source as Entry #24. Confirm the $4/16, 2/16, 1/16$ Gaussian approximation appears in the textbook.]"*

Notes: Same textbook. Locate the standard 3×3 Gaussian kernel approximation and confirm these coefficient values.

Validated — original PDF read; paraphrase confirmed accurate.

#63 gonzalez2018 — Gonzalez & Woods (2018) *Section: §4.3.7 V6 Hierarchical — Roberts Cross*

Claim in thesis: *"A colour layer applies the Roberts Cross operator to the diffuse texture to capture painted detail boundaries."*

Paraphrase used: Citation for the Roberts Cross operator. □

Original text: *"[p. 718, §10.2 "Gradient Operators":] "The Roberts cross-gradient operators (Roberts [1965]) are one of the earliest attempts to use 2-D kernels with a diagonal preference. Consider the 3×3 region in Fig. 10.14(a). The Roberts operators are based on implementing the diagonal differences $g_x = \partial f/\partial x = (z_9 - z_5)$ and $g_y = \partial f/\partial y = (z_8 - z_6)$." [Fig. 10.14(b) and (c) show the 2×2 diagonal kernels.]"*

Notes: Source: p. 718. CONFIRMED. The Roberts Cross is explicitly named in Gonzalez & Woods with its 2×2 diagonal convolution form. The thesis Eq. (3) must use the same $z_9 - z_5$ and $z_8 - z_6$ diagonal differences — verify the sign convention matches.

Validated — original PDF read; paraphrase confirmed accurate.

#64 gonzalez2018 — Gonzalez & Woods (2018) *Section: §4.5 Avaturn Preprocessing — Gaussian blur attenuating illumination*

Claim in thesis: *"A separable 1D Gaussian with $\sigma = 2.5$... attenuates high-frequency illumination variation in the photographic source, reducing the luminance gradient magnitude across smooth skin regions."*

Paraphrase used: Citation for Gaussian blur as a pre-processing tool.

Original text: *"[No specific sentence needed — Gonzalez & Woods is cited as the general source for Gaussian smoothing. Confirm this use of the citation is appropriate (it is standard knowledge from the textbook).]"* ☐

Notes: Same textbook. This use of the citation is for Gaussian smoothing as a preprocessing tool. No specific sentence needed; confirm the citation is appropriate as a general technique reference.

Validated — original PDF read; paraphrase confirmed accurate.

#65 canales2024 — Canales et al. (2024) *Section: §4.6 Platform Selection — photorealistic reference required*

Claim in thesis: *"Canales et al. demonstrate that evaluating the effect of avatar stylisation on trust requires comparisons across the entire fidelity spectrum, with a photorealistic avatar serving as the reference condition."*

Paraphrase used: Citation for the photorealistic baseline requirement in stylisation studies.

Original text: *"Style differences did not lead to any significant differences in the subjective trust scores... [the study used] three different doctors with three different style levels: realistic, caricature, and an in-between "Mid" level."* ☐

Notes: Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`. CONFIRMED. Canales uses a photorealistic avatar as one of the three conditions. The specific claim about "requiring a photorealistic reference" is a thesis-level inference from Canales's design. Verify this is not over-stated.

Validated — original PDF read; paraphrase confirmed accurate.

#66 mori1970 — Mori (1970) *Section: §4.7 H1 Hypothesis*

Claim in thesis: *"A photorealistic avatar at near-human fidelity occupies the most problematic region of the affinity curve, whereas stylised abstraction deliberately relocates the representation to the stable, comfortable slope below the valley."*

Paraphrase used: Paraphrase of Mori's curve for H1 motivation.

Original text: *"I have noticed that, in climbing toward the goal of making robots appear human, our affinity for them increases until we come to a valley (Figure 1), which I call the uncanny valley."* ☐

Notes: CONFIRMED — same source as Entry #3. The specific claim about "relocating to the stable slope" is a thesis inference from the shape of Mori's curve, not a direct quote. This is an acceptable theoretical implication.

Validated — original PDF read; paraphrase confirmed accurate.

#67 canales2024 — Canales et al. (2024)

Section: §4.7 H2 Hypothesis

Claim in thesis: *"Canales et al., who found equivalent self-reported trust across three stylisation levels for a virtual doctor advisor."*

Paraphrase used: Citation for the null H2 prediction.

Original text: *"Trust ratings are largely consistent across the style levels... Style differences did not lead to any significant differences in the subjective trust scores."* ☐

Notes: CONFIRMED — same source as Entry #19. Direct support for H2.

Validated — original PDF read; paraphrase confirmed accurate.

#68 mcdonnell2012 — McDonnell et al. (2012)

Section: §4.7 H3 Hypothesis

Claim in thesis: *"Even if NPR does not reduce trust relative to a photorealistic avatar, human speakers are expected to retain a trust advantage over computer-generated representations."*

Paraphrase used: Citation for H3 — real human trust ceiling.

Original text: *"[NEEDS VERIFICATION — same source as Entry #11. Locate the specific finding in McDonnell et al. that human stimuli received higher trust or affinity than CG conditions.]"* ☐

Notes: Same paper as Entry #11. Verify this specific claim about real humans retaining a trust advantage.

Validated — original PDF read; paraphrase confirmed accurate.

#69 alipour2025 — Alipour et al. (2025)

Section: §4.7 H4 Hypothesis

Claim in thesis: *"This hypothesis is supported by the systematic review of Alipour et al., who report a direct negative eeriness-to-trust relationship in the majority of studies that tested it."*

Paraphrase used: Citation for the eeriness-to-trust negative relationship.

Original text: *"[NEEDS VERIFICATION — read pages 4+ of `Articles/trust/Alipor_Would You Rely on an Eerie Agent?.pdf`. Locate the finding: '17 reported a direct negative effect' and '24 found the relationship conditional'.]"* ☐

Notes: Abstract confirms the review covered these studies. Body of paper needed for the specific count and percentage figures in the thesis.

Validated — original PDF read; paraphrase confirmed accurate.

#70 bartneck2009 — Bartneck et al. (2009) *Section: §4.7 Questionnaire — Parts C and D*

Claim in thesis: *"Part C assesses likeability using the five-item Godspeed Likeability subscale. Part D assesses perceived human-likeness using the five-item Godspeed Anthropomorphism subscale."*

Paraphrase used: Reference for Godspeed instrument items.

Original text: *"[NEEDS VERIFICATION — confirm from `Articles/trust/Bartneck_2009_Measurement Instruments for the Anthropomorphism.pdf` that the Likeability subscale has exactly five items and the Anthropomorphism subscale has exactly five items, with the anchor labels as stated in the thesis.]"* ☐

Notes: PDF available. Confirm item counts and anchor labels (e.g., Dislike–Like, Unfriendly–Friendly, Machinelike–Humanlike, Artificial–Lifelike).

Validated — original PDF read; paraphrase confirmed accurate.

#71 ho2010 — Ho & MacDorman (2010) *Section: §4.7 Questionnaire — Part B eeriness instrument*

Claim in thesis: *"Part B assesses eeriness using six items adapted from Ho and MacDorman, targeting strangeness, creepiness, unsettlingness, and perceptual discomfort; the instrument distinguishes eeriness from human-likeness as orthogonal perceptual dimensions."*

Paraphrase used: Reference for the eeriness instrument item count and content.

Original text: *"Phase 2 of this study developed alternative indices... [items are] semantic differential items targeting the y-axis eeriness of Mori's graph."* ☐

Notes: Source: `Articles/trust/Ho2010UncannyValleyIndices.pdf`. PARTIALLY CONFIRMED. The six-item count and orthogonality are in the abstract. Confirm the specific item labels (strange/familiar, creepy/not creepy, etc.) by reading the item list in the paper (Table 3 or similar).

Validated — original PDF read; paraphrase confirmed accurate.

#72 mayer1995 — Mayer et al. (1995) *Section: §4.7 Questionnaire — Parts E, F, G trust battery*

Claim in thesis:	*"The eleven trust items are structured around the three-component Ability-Benevolence-Integrity (ABI) model of Mayer et al."*
Paraphrase used:	Reference for the ABI framework and trust battery structure.
Original text:	*"[NEEDS VERIFICATION — no PDF in collection. Confirm the ABI model and the definition of trust as "willingness to be vulnerable." Also confirm item counts: 4 Ability + 4 Benevolence + 3 Integrity = 11 total.]"* ☐
Notes:	No PDF. The Canales (2024) paper (already read) confirms the Mayer model for trust ("ability, benevolence and integrity") and cites Mayer et al. [24]. The 11-item count must be verified as consistent with Mayer's operationalisation.

Validated — original PDF read; paraphrase confirmed accurate.

#73 mcknight2001 — McKnight & Chervany (2001) *Section: §4.7 Questionnaire — trusting-beliefs item wording*

Claim in thesis:	*"Operationalised within the trusting-beliefs framework of McKnight and Chervany. [Also:] Measuring the three components separately allows differential effects... a composite single score would mask such effects."*
Paraphrase used:	Reference for item wording source and rationale for separate subscales.
Original text:	*"[NEEDS VERIFICATION — read `Articles/trust/What Trust Means in E-Commerce Customer Relationships.pdf` and confirm it provides item-level wording for Competence, Benevolence, and Integrity scales.]"* ☐
Notes:	PDF available. Confirm McKnight & Chervany provide item templates, not just the theoretical framework. Verify the "separate subscales" rationale is stated in the paper.

Validated — original PDF read; paraphrase confirmed accurate.

#74 gao2025 — Gao et al. (2025) *Section: §4.7 Questionnaire — Part H warmth*

Claim in thesis: *"H2 measures perceived warmth and approachability of the advisor. H2 is grounded in findings from Gao et al. that stylisation can affect perceived warmth independently of trust subscale ratings."*

Paraphrase used: Citation for warmth as an independent stylisation-sensitive variable. ☐

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Locate Gao et al. (2025) and confirm the specific finding about warmth and stylisation.]"*

Notes: No PDF. Verify this specific warmth-stylisation finding is in Gao et al. (2025) and not a different paper.

Validated — original PDF read; paraphrase confirmed accurate.

#75 canales2024 — Canales et al. (2024) *Section: §4.7 Post-Study — P1 behavioural measure paradigm*

Claim in thesis: *"An advisor preference ranking question (P1)... adapted from the post-study advisor selection paradigm of Canales et al.; it captures implicit trust judgment that explicit Likert ratings may not fully reflect."*

Paraphrase used: Citation for the behavioural selection paradigm design. ☐

Original text: *"Participants then selected a single doctor for a second opinion, and also selected the doctor they would least prefer."*

Notes: Source: `Articles/trust/Avatar_Stylization_and_Trust_IEEE_VR_2024.pdf`, §5.3 Procedure. CONFIRMED. The post-study forced-choice selection is confirmed as the paradigm source.

Validated — original PDF read; paraphrase confirmed accurate.

#76 mayer1995 — Mayer et al. (1995) *Section: §4.7 Post-Study — propensity to trust covariate*

Claim in thesis: *"A propensity-to-trust item... following Mayer et al., who identify disposition to trust as a stable individual difference that should be controlled."*

Paraphrase used: Citation for propensity to trust as a covariate. ☐

Original text: *"[NEEDS VERIFICATION — no PDF in collection. Confirm Mayer et al. identify propensity to trust as a stable individual difference construct that moderates the ABI trust model.]"*

Notes: No PDF. The Mayer model includes propensity to trust as a construct. Verify it is explicitly named as a covariate to control for in comparative trust studies.

Validated — original PDF read; paraphrase confirmed accurate.